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# RURALNET TRAINING COLLEGE

## ICT ESSENTIALS

## COMPREHENSIVE GUIDE FOR STUDENTS

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# WEEK ONE:

# INTRODUCTION TO COMPUTERS AND COMPUTER COMPONENTS

## TOPIC 1: INTRODUCTION TO COMPUTERS

### 1.1 Definition of a Computer

A computer is an electronic programmable device that accepts raw data as input, processes the data according to a set of instructions, stores data and information, and produces meaningful output.

A computer performs four major operations:

1. **Input:** Receiving data and instructions from the user.
2. **Processing:** Manipulating the data according to given instructions.
3. **Output:** Presenting the processed data in a meaningful form.
4. **Storage:** Saving data and information for future use.

**Example**

Consider a student using a computer to calculate examination results.

- Input: The student enters marks obtained in different subjects.
- Processing: The computer calculates the total and average marks.
- Output: The computer displays the total and average.
- Storage: The results are saved for future reference.

### 1.2 The Information Processing Cycle

The information processing cycle describes how a computer transforms raw data into useful information.

The four major stages are:

**Input → Processing → Output → Storage**

#### 1. Input

Input involves entering raw data and instructions into a computer using input devices such as a keyboard, mouse, scanner or microphone.

#### 2. Processing

Processing involves manipulating the entered data to produce meaningful information. This is mainly performed by the Central Processing Unit (CPU).

#### 3. Output

Output is the presentation of processed information to the user. It can be displayed on a monitor, printed on paper or played through speakers.

#### 4. Storage

Storage involves keeping data, instructions and information for future use.

**Example:**

When preparing a student's school fees statement:

- Input: Student name, registration number, fees payable and amount paid.
- Processing: Calculate the outstanding balance.
- Output: Display the student's fees statement.
- Storage: Save the statement in the computer.

## 1.3 Basic Components of a Computer System

A computer system depends on several components working together.

### A. Hardware

These are the physical parts of a computer that can be seen and physically touched.

Examples:

- Monitor
- Keyboard
- Mouse
- System unit
- Printer
- Speakers
- Hard disk

### B. Software

Software refers to programs and instructions that tell computer hardware what to do and how to perform different tasks.

Examples:

- Windows 11
- Microsoft Word
- Google Chrome
- Microsoft Excel
- Adobe Photoshop

### C. Peopleware (Liveware)

Peopleware refers to the human beings who interact with, operate, manage or maintain computer systems.

Examples:

- Computer users
- Computer operators
- Programmers
- Network administrators
- Computer technicians
- Data entry clerks

### D. Procedures

Procedures are the established rules and instructions followed when operating computer systems.

Examples:

- Procedures for starting and shutting down computers.
- Procedures for saving documents.
- Procedures for backing up information.
- Procedures for protecting computer systems.

# TOPIC 2: BASIC COMPUTER TERMINOLOGIES

Students should understand the following terms before proceeding to advanced computer operations.

## 2.1 Data

Data refers to raw facts and figures that have not yet been processed into meaningful information.
Examples:

- 500
- John

Individually, these values may not communicate a complete message.

## 2.2 Information

Information refers to processed and organised data that has meaning and can be used for decision-making.
Example:
If the numbers 500, 750 and 1,000 represent amounts paid by three students, arranging and analysing them provides useful financial information.

## 2.3 Processing

Processing is the manipulation of raw data according to instructions to produce useful information.
Examples include:

- Calculating totals.
- Sorting student records.
- Comparing examination marks.
- Preparing financial reports.

## 2.4 Program

A program is a set of instructions written to enable a computer to perform a particular task.
Examples:

- Microsoft Word
- Microsoft Excel
- Google Chrome

## 2.5 Peripheral Devices

Peripheral devices are additional hardware components connected to a computer to provide input, output, storage or communication functions.
Examples include printers, scanners, external drives and webcams.

## 2.6 Port

A port is a physical connection point or interface used to connect external devices to a computer.
Common computer ports include:

1. **USB port:** Connects flash drives, keyboards, mice, phones and other USB devices.
2. **HDMI port:** Transmits digital video and audio to monitors, televisions and projectors.
3. **Ethernet port (RJ45):** Connects a computer to a wired network.
4. **Audio ports:** Connect headphones, microphones and external speakers.
5. **DisplayPort:** Transmits digital video and audio to compatible displays.
6. **USB-C port:** Supports data transfer, charging and, on compatible devices, video output.
7. **VGA port:** An older analogue video connection found on some older computers and projectors.

**Note:** Older interfaces such as PS/2, serial and parallel ports may still be found on legacy equipment but are less common on modern computers.

## 2.7 Booting

Booting is the process of starting a computer and loading the operating system into memory so that the computer becomes ready for use.

## 2.8 Operating System

An operating system is system software that manages computer hardware and software resources and provides an environment through which users interact with the computer.
Examples include Windows, Linux, macOS, Android and iOS.

## 2.9 File

A file is a collection of related information stored under a particular name on a storage device.
Examples:

- Student.docx
- Results.xlsx
- Photo.jpg
- Introduction.pdf

## 2.10 Folder

A folder is a digital container used to organise and store files and other folders.

## 2.11 Desktop

The desktop is the main working area displayed after a computer finishes loading its operating system and the user signs in.

## 2.12 Icon

An icon is a small graphical representation of a program, file, folder or system function.

### 2.13 Cursor and Pointer

- **Pointer:** The graphical indicator controlled by a mouse or other pointing device.
- **Text cursor:** The blinking vertical line that indicates where typed characters will appear.

### 2.14 User Interface (UI)

A user interface is the means through which a person interacts with a computer or software application.
Two common types are:

- Graphical User Interface (GUI).
- Command-Line Interface (CLI).

## TOPIC 3: CHARACTERISTICS OF COMPUTERS

Computers have several characteristics that make them useful in modern society.

### 3.1 Speed

Computers process large amounts of data within very short periods.
For example, a computer can calculate examination results for hundreds of students within seconds.

### 3.2 Accuracy

Computers produce accurate results when the data entered and instructions provided are correct.
Incorrect data or instructions can, however, produce incorrect results.
This is commonly described as Garbage In, Garbage Out (GIGO).

### 3.3 Storage Capacity

Computers can store large amounts of data, including documents, photographs, videos, databases and software.

### 3.4 Reliability

Computers can perform repetitive operations consistently without becoming physically tired.

### 3.5 Automation

Once instructed and properly configured, computers can perform tasks automatically with limited human intervention.

### 3.6 Versatility

Computers can perform many different tasks, including typing, calculations, graphic design, communication, programming and entertainment.

### 3.7 Diligence

Computers can perform repetitive tasks for long periods without losing concentration due to fatigue.

### 3.8 Multitasking

Modern computers can run multiple applications and processes at the same time.
For example, a user can listen to music, download a document and type a report simultaneously.

# TOPIC 4: CLASSIFICATION AND TYPES OF COMPUTERS

Computers can be classified according to their size, processing capabilities, functionality and purpose.

## 4.1 Classification According to Size and Processing Capacity

### A. Supercomputers

Supercomputers are extremely powerful computers designed to perform highly complex calculations and process enormous amounts of data.

Applications:

- Weather forecasting.
- Scientific research.
- Climate modelling.
- Space exploration.
- Advanced artificial intelligence research.
- Complex engineering simulations.

### B. Mainframe Computers

Mainframe computers are large, powerful systems designed to handle huge volumes of data and support many users and transactions simultaneously.

Applications:

- Banking systems.
- Government databases.
- Airline reservation systems.
- Large insurance companies.
- Large-scale business operations.

### C. Minicomputers

Minicomputers were traditionally medium-sized multi-user computers, smaller than mainframes but more powerful than many early microcomputers.

They were commonly used by medium-sized organisations and institutions.

**Modern context:** The term is largely historical. Many functions previously handled by minicomputers are now performed by servers and other enterprise systems.

### D. Microcomputers

Microcomputers are computers designed primarily for individual users.

Examples:

1. Desktop computers.
2. Laptop computers.
3. Tablet computers.
4. Smartphones.
5. Workstations.
6. Small single-board computers.

## 4.2 Classification According to Functionality

### A. Analogue Computers

Analogue computers process information using continuously varying physical quantities such as temperature, pressure, speed and voltage.

Examples:

- Traditional analogue flight instruments.

- Analogue measuring and control systems.
- Some specialised scientific instruments.

### B. Digital Computers

Digital computers process information using discrete values, primarily represented in binary form (0 and 1).

Examples:

- Desktop computers.
- Laptops.
- Smartphones.
- Digital calculators.

### C. Hybrid Computers

Hybrid computers combine features of analogue and digital computing systems.

They can receive continuously varying physical measurements and convert or process them digitally.

Applications:

- Medical monitoring systems.
- Industrial control systems.
- Scientific measurement systems.

## 4.3 Classification According to Purpose

### A. General-Purpose Computers

These are designed to perform many different tasks depending on the software installed.

Examples:

- Desktop computers.
- Laptops.
- Tablets.

### B. Special-Purpose Computers

These are designed to perform a specific task or a limited range of tasks.

Examples:

- ATM control systems.
- Traffic light controllers.
- Digital vehicle control systems.
- Dedicated industrial control computers.

# TOPIC 5: APPLICATION AREAS OF COMPUTERS

Computers are used in almost every sector of modern society.

## 5.1 Education

Computers support:

- Online learning.
- Preparation of examination papers.
- Research and assignments.
- Student records management.
- Virtual classrooms.
- Digital presentations.
- Computer-based examinations.

## 5.2 Business and Commerce

Computers are used for:

- Accounting and bookkeeping.
- Payroll processing.
- Stock management.
- Customer records.
- Sales and purchases.
- Business communication.
- Point of Sale (POS) operations.

## 5.3 Banking and Finance

Applications include:

- Electronic fund transfers.
- Mobile banking.
- ATM services.
- Account management.
- Financial reporting.
- Fraud detection.

## 5.4 Healthcare

Computers are used for:

- Patient record management.
- Medical imaging.
- Laboratory analysis.
- Hospital management.
- Telemedicine.
- Patient monitoring.

## 5.5 Communication

Computers support:

- Email communication.
- Video conferencing.
- Instant messaging.
- Social networking.
- Internet browsing.

## 5.6 Government

Applications include:

- Tax administration.
- National registration systems.
- Digital public services.
- Government records management.
- Revenue collection.

## 5.7 Engineering and Manufacturing

Computers support:

- Computer-Aided Design (CAD).
- Computer-Aided Manufacturing (CAM).
- Robotics.
- Product modelling.

- Industrial automation.

### 5.8 Entertainment and Media

Computers are used for:

- Video editing.
- Music production.
- Gaming.
- Animation.
- Graphic design.
- Film production.
- Digital photography.

### 5.9 Agriculture

Applications include:

- Weather monitoring.
- Farm management.
- Irrigation control.
- Crop monitoring.
- Agricultural research.
- Market information.

### 5.10 Transport and Logistics

Computers support:

- GPS navigation.
- Vehicle tracking.
- Airline reservations.
- Traffic management.
- Fleet management.
- Online ticketing.

### 5.11 Artificial Intelligence

Computers are used in artificial intelligence applications such as:

- Chatbots.
- Language translation.
- Speech recognition.
- Image recognition.
- Predictive analysis.
- Automated decision-support systems.

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## TOPIC 6: ADVANTAGES AND DISADVANTAGES OF COMPUTERS

### 6.1 Advantages

1. **Speed:** Computers process information quickly.
2. **Accuracy:** They produce accurate results when correctly instructed.
3. **Large storage capacity:** They store large volumes of information.
4. **Communication:** They support instant communication across geographical locations.
5. **Automation:** They automate repetitive and complex tasks.
6. **Research:** They provide access to educational and research resources.
7. **Productivity:** They make work easier and faster.

8. **Entertainment:** They support music, games, videos and other forms of entertainment.
9. **Data analysis:** They help organisations analyse information for decision-making.
10. **Accessibility:** Assistive technologies enable people with different disabilities to use computers.

## 6.2 Disadvantages

1. **Health problems:** Prolonged use may cause eye strain, back pain and repetitive strain injuries.
2. **Cybersecurity threats:** Computers may be attacked by viruses, ransomware and other malicious software.
3. **Privacy concerns:** Personal information may be exposed or misused.
4. **Unemployment:** Automation may reduce the demand for certain types of manual work.
5. **Distraction:** Games and social media can interfere with productivity.
6. **High maintenance costs:** Hardware repairs, software licences and upgrades can be expensive.
7. **Dependence:** Excessive reliance on computers may reduce some manual skills.
8. **Electronic waste:** Improper disposal of computers can damage the environment.
9. **Technical failures:** Hardware and software problems may interrupt work.
10. **Digital divide:** Unequal access to computers and the internet creates differences in opportunities.

# TOPIC 7: COMPUTER HARDWARE

## 7.1 Definition of Computer Hardware

Computer hardware refers to all the physical components of a computer system that can be seen and touched.

Hardware is broadly classified into:

1. Input devices.
2. Output devices.
3. Storage devices.
4. Processing devices.
5. Communication devices.

## 7.2 The System Unit

The system unit is the main computer enclosure that houses important internal hardware components.

It is commonly referred to as the computer case or CPU casing, although the CPU itself is only one component inside it.

### Main Components Found Inside the System Unit

#### A. Motherboard

The motherboard is the main circuit board that connects and allows communication between different computer components.

It provides connections for:

- Processor.
- RAM.
- Storage devices.
- Expansion cards.
- Power connections.
- External ports.

**B. Central Processing Unit (CPU)**
The CPU executes program instructions and performs calculations.
**C. Random Access Memory (RAM)**
RAM temporarily holds data and instructions currently being used by the computer.
**D. Storage Drive**
This stores the operating system, applications, documents and other files.
Examples:

- HDD.
- SATA SSD.
- NVMe SSD.

**E. Power Supply Unit (PSU)**
The PSU converts electrical power from the mains supply into suitable voltages required by the computer's internal components.
**F. Cooling System**
The cooling system prevents computer components from overheating.
It may include:

- Cooling fans.
- Heat sinks.
- Liquid cooling systems.

**G. Graphics Processing Unit (GPU)**
The GPU processes graphics and visual information.
It may be integrated into the processor or installed as a separate graphics card.
**H. Network Interface Card (NIC)**
A network interface allows the computer to connect to a network through Ethernet or wireless communication.

### 7.3 External Computer Components

Common external components include:

- Monitor.
- Keyboard.
- Mouse.
- Speakers.
- Webcam.
- Microphone.
- Printer.
- Scanner.
- External storage devices.

# TOPIC 8: COMPUTER INPUT DEVICES

### 8.1 Definition

Input devices are hardware components used to enter data and instructions into a computer.
They enable users to communicate with the computer.

### 8.2 Examples of Input Devices

#### A. Keyboard

A keyboard is an input device used to enter letters, numbers, symbols and commands into a computer.
It is discussed in detail under Topic 15.

## B. Mouse

A mouse is a pointing device used to control the pointer on a computer screen.
It is discussed in detail under Topic 16.

## C. Scanner

A scanner captures printed documents, photographs and other physical materials and converts them into digital form.
Types include:

- Flatbed scanner.
- Sheet-fed scanner.
- Handheld scanner.
- Portable document scanner.

**Procedure for scanning a document:**

1. Connect the scanner to the computer.
2. Switch on the scanner.
3. Place the document on the scanner glass or feed it into the automatic document feeder.
4. Open the scanning application.
5. Select the scanner.
6. Choose the required colour mode and resolution.
7. Preview the document if the application provides that option.
8. Click Scan.
9. Select the storage location.
10. Save the scanned document in the required format, such as PDF or JPEG.

## D. Microphone

A microphone captures sound and converts it into electronic signals that can be processed by a computer.
Uses:

- Voice recording.
- Online meetings.
- Voice commands.
- Podcast production.
- Audio editing.

## E. Webcam

A webcam captures still images and live video.
Uses:

- Video conferencing.
- Online classes.
- Video recording.
- Security monitoring.

## F. Touchscreen

A touchscreen allows users to interact with a computer by touching items displayed on the screen.
It can perform the functions of both input and output devices.
Examples:

- Smartphones.
- Tablets.
- Touch-enabled laptops.

- Information kiosks.

### G. Barcode Reader

A barcode reader scans and interprets information encoded in barcodes.
It is commonly used in:

- Supermarkets.
- Retail shops.
- Warehouses.
- Libraries.
- Inventory management.

### H. QR Code Scanner

A QR code scanner reads two-dimensional codes containing information such as website addresses, payment details and product information.
Modern smartphones can scan QR codes using their cameras.

### I. Graphics Tablet

A graphics tablet is an input device used with a stylus to draw, write and create digital artwork.
It is commonly used by graphic designers and digital illustrators.

### J. Biometric Scanner

A biometric scanner captures unique physical characteristics for identification or authentication.
Examples:

- Fingerprint scanners.
- Facial recognition cameras.
- Iris scanners.

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# TOPIC 9: COMPUTER OUTPUT DEVICES

## 9.1 Definition

Output devices are hardware components used to present processed information to users.
They convert computer-processed information into forms that people can understand.

## 9.2 Examples of Output Devices

### A. Monitor

A monitor is an electronic display device used to present text, images, videos and other visual information.
Common types:

- LCD monitors.
- LED-backlit monitors.
- OLED displays.

Important monitor specifications include:

- Screen size.
- Resolution.
- Refresh rate.
- Display technology.

### B. Printer

A printer produces hard-copy output by transferring digital information onto paper or other printable materials.

**Types of Printers**

**1. Inkjet Printer**

Uses liquid ink sprayed through tiny nozzles onto paper.
Commonly used for:

- Colour photographs.
- Home printing.
- Office documents.

**2. Laser Printer**
Uses laser technology and toner powder to produce printed documents.
Advantages:

- High printing speed.
- Good text quality.
- Suitable for high-volume printing.

**3. Dot Matrix Printer**
Uses pins that strike an ink ribbon to produce characters and images.
It is commonly associated with older systems and multipart forms.
**Procedure for Printing a Document**

1. Connect the printer to the computer through USB or a supported network connection.
2. Switch on the printer.
3. Ensure that the printer has paper and sufficient ink or toner.
4. Open the document you want to print.
5. Click File.
6. Select Print.
7. Choose the correct printer.
8. Specify the number of copies.
9. Select the required pages.
10. Adjust orientation and paper size if necessary.
11. Click Print.

## C. Speakers

Speakers convert audio signals into sound.
Uses:

- Listening to music.
- Watching videos.
- Attending online meetings.
- Listening to system notifications.

## D. Headphones

Headphones provide personal audio output.
Some headphones also include microphones for communication.

## E. Projector

A projector displays computer images and videos on a large screen or wall.
Common applications:

- Classroom teaching.
- Business presentations.
- Meetings.
- Training sessions.

## F. Plotter

A plotter produces large and detailed drawings, particularly technical designs.
It is commonly used in:

- Architecture.
- Engineering.
- Construction.
- Technical drawing.

### 9.3 Soft Copy and Hard Copy

**Soft Copy:** Information presented or stored in digital form.
Examples:

- PDF documents.
- Word documents.
- Digital photographs.

**Hard Copy:** Information produced in physical form, usually on paper.
Examples:

- Printed examination results.
- Printed reports.
- Receipts.

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# TOPIC 10: COMPUTER STORAGE DEVICES

## 10.1 Definition of Computer Storage

Computer storage refers to the technology used to retain digital data, programs and information for temporary or long-term use.
Storage is important because it enables users to save their work and retrieve it later.

## 10.2 Units of Computer Storage

Computers store information using binary digits.
The following are common storage units:

| Unit | Description |
|---|---|
| Bit | Smallest unit of digital data; 0 or 1 |
| Nibble | 4 bits |
| Byte | 8 bits |
| Kilobyte (KB) | Approximately 1,000 bytes |
| Megabyte (MB) | Approximately 1,000 KB |
| Gigabyte (GB) | Approximately 1,000 MB |
| Terabyte (TB) | Approximately 1,000 GB |
| Petabyte (PB) | Approximately 1,000 TB |

**Note:** Computers and storage manufacturers commonly use decimal measurements, while some operating systems use binary measurements. Therefore, displayed storage sizes may differ slightly.

## 10.3 Classification of Computer Memory

Computer memory and storage can be grouped into primary memory and secondary storage.

### A. Primary Memory

Primary memory is directly accessible to the processor and is used to hold data and instructions required during computer operations.

**1. Random Access Memory (RAM)**

RAM is temporary working memory used to store data and instructions that are currently being processed.
Characteristics:

- It is generally volatile.
- Its contents are lost when power is switched off.
- It allows fast access to active data.
- Its capacity affects how many applications can run comfortably.

Types of RAM:

**a. Dynamic RAM (DRAM)**

DRAM stores information in memory cells that require periodic refreshing.
It is commonly used as the main system memory.

**b. Static RAM (SRAM)**

SRAM stores information without requiring periodic refreshing while power is available.
It is faster and more expensive per unit of capacity than DRAM and is commonly used in processor cache memory.
Modern RAM generations include DDR3, DDR4 and DDR5. DDR6 and later technologies should be understood according to their actual availability and adoption rather than assumed to be present in every computer.

**2. Read Only Memory (ROM)**

ROM is non-volatile memory used to store instructions and firmware required for the operation of electronic devices.
Unlike RAM, it retains information when power is switched off.
Types include:

- **MROM:** Mask Read Only Memory, programmed during manufacturing.
- **PROM:** Programmable Read Only Memory, which can be programmed once.
- **EPROM:** Erasable Programmable Read Only Memory, which can be erased using ultraviolet light.
- **EEPROM:** Electrically Erasable Programmable Read Only Memory.

Modern computer firmware is commonly stored in flash memory, a form of non-volatile memory related to EEPROM technology.

## B. Secondary Storage

Secondary storage refers to devices used to retain data and information for long periods, even when the computer is switched off.
Examples:

**1. Hard Disk Drive (HDD)**

An HDD stores information on rotating magnetic disks.
Advantages:

- Relatively affordable for large storage capacities.
- Suitable for storing large files.

Disadvantages:

- Contains moving mechanical components.
- Generally slower than SSDs.
- More vulnerable to physical shock.

**2. Solid State Drive (SSD)**

An SSD stores information using flash memory and has no moving mechanical parts.
Advantages:

- Faster than traditional HDDs.
- Silent operation.
- Lower power consumption in many applications.

- More resistant to physical shock.

Types:

- SATA SSD.
- NVMe SSD.

**3. USB Flash Drive**

A portable storage device that connects through a USB port.

Uses:

- Transferring documents.
- Sharing photographs.
- Carrying presentations.
- Installing software and operating systems.

**4. Memory Card**

A small removable storage device commonly used in:

- Smartphones.
- Cameras.
- Tablets.
- Other portable devices.

Examples include SD and microSD cards.

**5. Optical Discs**

These use laser technology to read and write information.

Examples:

- CD.
- DVD.
- Blu-ray Disc.

Older formats such as floppy disks and magnetic tapes may be encountered in historical or specialised systems, but they are not standard everyday storage options on modern personal computers.

**6. Cloud Storage**

Cloud storage allows users to store files on remote servers accessed through the internet.

Examples:

- Google Drive.
- Microsoft OneDrive.
- Dropbox.
- iCloud.

Advantages:

- Access files from different devices.
- Easier file sharing.
- Supports online backup and collaboration.
- Reduces dependence on one physical storage device.

## 10.4 Differences Between RAM and ROM

| RAM | ROM |
|---|---|
| Random Access Memory | Read Only Memory |
| Generally volatile | Non-volatile |
| Temporarily stores active data | Stores firmware and essential instructions |
| Can be read and written during normal operation | Writing depends on the ROM technology |
| Contents are normally lost when power is removed | Retains contents without power |

### 10.5 Differences Between HDD and SSD

| HDD | SSD |
|---|---|
| Uses rotating magnetic disks | Uses flash memory |
| Has moving parts | Has no moving mechanical parts |
| Generally slower | Generally faster |
| Often cheaper per GB | Often more expensive per GB |
| More sensitive to physical shock | Generally more resistant to physical shock |

# TOPIC 11: THE CENTRAL PROCESSING UNIT (CPU)

## 11.1 Definition

The Central Processing Unit (CPU), commonly called the processor, is the main processing component of a computer.
It interprets and executes instructions, performs calculations and controls many operations within the computer system.
It is commonly described as the brain of the computer.
**Important:** The CPU is a processor chip, not the entire system unit or computer casing.

## 11.2 Main Functions of the CPU

1. Executes program instructions.
2. Performs arithmetic calculations.
3. Performs logical comparisons.
4. Controls the flow of information.
5. Coordinates activities between different computer components.

## 11.3 Main Components of the CPU

### A. Control Unit (CU)

The Control Unit directs and coordinates computer operations.
Functions:

- Fetches instructions.
- Decodes instructions.
- Controls execution.
- Coordinates communication between computer components.

### B. Arithmetic and Logic Unit (ALU)

The ALU performs arithmetic and logical operations.
Arithmetic operations include:

- Addition.
- Subtraction.
- Multiplication.
- Division.

Logical operations include:

- Greater than.
- Less than.
- Equal to.
- Logical AND.
- Logical OR.
- Logical NOT.

### C. Registers
Registers are very small, high-speed storage locations inside the processor.
They temporarily hold data, instructions and addresses required during processing.

### D. Cache Memory
Cache is high-speed memory located inside or close to the processor.
It stores frequently used data and instructions to reduce the time required to access information from slower memory.

## 11.4 The CPU Processing Cycle
The CPU commonly follows the following instruction cycle:

1. **Fetch:** Retrieves an instruction from memory.
2. **Decode:** Interprets the instruction to determine what must be done.
3. **Execute:** Performs the required operation.
4. **Store:** Saves the result where required.

This is commonly referred to as the Fetch–Decode–Execute cycle.

## 11.5 CPU Performance
CPU performance is influenced by several factors:

- Clock speed.
- Number of processing cores.
- Processor architecture.
- Cache size.
- Power consumption.
- Cooling and operating temperature.

Examples of processor manufacturers include Intel, AMD and Apple.

# TOPIC 12: COMPUTER SOFTWARE
## 12.1 Definition
Computer software refers to the programs, applications and instructions that enable computer hardware to perform specific tasks.
Software cannot be physically touched like hardware.

## 12.2 Classification of Computer Software
Computer software is broadly classified into:

1. System software.
2. Application software.

### A. System Software
System software manages computer hardware and provides the environment required for application software to operate.
Types include:

**1. Operating Systems**
An operating system manages computer resources and provides an interface between the user, applications and hardware.
Examples:

- Windows 10.
- Windows 11.
- macOS.
- Linux.

- Android.
- iOS.

**2. Device Drivers**

Device drivers are software components that allow the operating system to communicate with hardware devices.

Examples:

- Printer drivers.
- Graphics drivers.
- Audio drivers.
- Network adapter drivers.

**3. Utility Software**

Utility software helps maintain, protect and optimise computer systems.

Examples:

- Antivirus software.
- Backup utilities.
- Disk management tools.
- File compression tools.
- Storage cleanup tools.

**4. Firmware**

Firmware is software stored in non-volatile memory that provides low-level control of hardware devices.

Examples:

- BIOS/UEFI firmware.
- Router firmware.
- Printer firmware.

## B. Application Software

Application software consists of programs designed to help users perform specific tasks.

Examples:

| Application | Main Purpose |
|---|---|
| Microsoft Word | Word processing |
| Microsoft Excel | Spreadsheets and calculations |
| Microsoft PowerPoint | Presentations |
| Microsoft Access | Database management |
| Google Chrome | Web browsing |
| Adobe Photoshop | Image editing |
| CorelDRAW | Graphic design |
| VLC Media Player | Playing multimedia |
| QuickBooks | Accounting |
| Zoom | Online meetings |

## 12.3 Types of Application Software

**1. General-Purpose Applications**

These are designed to perform common tasks.

Examples:

- Word processors.
- Spreadsheet applications.
- Presentation applications.

- Web browsers.

**2. Special-Purpose Applications**

These are developed to perform particular tasks or serve specific industries.

Examples:

- Payroll systems.
- School management systems.
- Hospital management systems.
- Point of Sale systems.
- Inventory management systems.

**3. Customised Software**

This is software developed or modified to meet the specific needs of an individual or organisation.

Example: A college student management system developed specifically for RuralNet Training College.

### 12.4 Differences Between Hardware and Software

| Hardware | Software |
|---|---|
| Physical components | Programs and instructions |
| Can be physically touched | Cannot be physically touched |
| Can suffer physical damage | Can become corrupted or infected |
| Manufactured | Developed or programmed |
| Examples: keyboard, monitor | Examples: Windows, Word |

# WEEK TWO: COMPUTER OPERATIONS AND PRACTICAL SKILLS

# TOPIC 13: OPERATING SYSTEMS

### 13.1 Definition

An operating system (OS) is system software that manages computer hardware and software resources and provides an environment for users to interact with the computer.

It acts as an intermediary between the user, applications and hardware.

## 13.2 Examples of Operating Systems

**Desktop and Laptop Operating Systems**

- Microsoft Windows.
- macOS.
- Linux distributions such as Ubuntu and Fedora.

**Mobile Operating Systems**

- Android.
- iOS.

## 13.3 Functions of an Operating System

**1. Memory Management**

The operating system allocates and manages computer memory for running programs.

**2. Process Management**

It manages running programs and determines how processor resources are shared among them.

**3. File Management**

It organises files and folders and controls access to stored information.

### 4. Device Management

It manages connected hardware such as printers, keyboards, storage drives and network adapters.

### 5. User Interface

It provides a means through which users interact with the computer.

### 6. Security Management

It helps protect computer resources through user accounts, permissions and security controls.

### 7. Resource Management

It coordinates the use of CPU time, memory, storage and other system resources.

### 8. Network Management

It supports network connections and communication with other computers and devices.

## 13.4 Types of User Interfaces

### A. Graphical User Interface (GUI)

A GUI allows users to interact with a computer through graphical elements such as:

- Windows.
- Icons.
- Menus.
- Buttons.
- Pointer.

Examples: Windows and macOS.

### B. Command-Line Interface (CLI)

A CLI allows users to interact with a computer by typing text commands.

Examples:

- Windows Command Prompt.
- Windows PowerShell.
- Linux Terminal.

# TOPIC 14: STARTING, RESTARTING AND SHUTTING DOWN A COMPUTER

## 14.1 Booting

Booting is the process through which a computer starts and loads the operating system into memory.

## 14.2 Types of Booting

### A. Cold Booting

Cold booting refers to starting a computer from a completely powered-off state.

**Procedure for cold booting:**

1. Ensure the computer is properly connected to a power source.
2. Check that the monitor, keyboard and other required peripherals are connected.
3. Switch on the power supply where applicable.
4. Press the computer's power button.
5. Allow the computer to perform its initial hardware checks.
6. Wait for the operating system to load.
7. Sign in using the authorised user account.

8. Wait for the desktop to become ready.

During startup, the computer performs a process called Power-On Self-Test (POST).
POST checks essential hardware components before the operating system loads.
Modern computers commonly use UEFI firmware, while older computers may use traditional BIOS firmware.

### B. Warm Booting

Warm booting refers to restarting a computer without completely switching off its power.

**Procedure for restarting Windows 10/11:**

1. Save all open documents.
2. Close applications where possible.
3. Click the Start button.
4. Select the Power icon.
5. Click Restart.
6. Wait for the computer to shut down and start again.

**Alternative method:**

Press `Ctrl + Alt + Delete`, select the Power icon and click Restart.

## 14.3 Sleep Mode

Sleep mode is a low-power state that allows the computer to resume work quickly.
It keeps the active session in memory while using reduced power.

**Procedure:**

1. Save your work.
2. Click Start.
3. Click the Power icon.
4. Select Sleep.

## 14.4 Hibernate Mode

Hibernate saves the current computer session to storage and then powers down the computer.
It consumes little or no power while the computer is hibernating.

## 14.5 Shutting Down a Computer

Shutting down properly allows the operating system to close applications and complete essential system operations.

**Procedure:**

1. Save all open documents.
2. Close applications.
3. Click Start.
4. Select the Power icon.
5. Click Shut down.
6. Wait for the computer to switch off completely.

**Important:** Avoid switching off a computer directly from the power socket unless there is an emergency. Sudden power interruption may result in unsaved work or system problems.

---

# TOPIC 15: THE COMPUTER KEYBOARD

## 15.1 Definition

A keyboard is an input device used to enter text, numbers, symbols and commands into a computer.

It is one of the most commonly used computer input devices.

## 15.2 Types of Keyboards

1. Standard wired keyboard.
2. Wireless keyboard.
3. Mechanical keyboard.
4. Ergonomic keyboard.
5. Multimedia keyboard.
6. Virtual or on-screen keyboard.

## 15.3 Main Sections of a Keyboard

### A. Function Keys

These are usually labelled F1 to F12 and are located along the upper section of a standard keyboard.
Their functions depend on the operating system and application being used.
Examples:

- F1: Commonly opens Help.
- F2: Renames a selected file or folder in Windows.
- F5: Refreshes a window or webpage in many applications.
- F11: Commonly toggles full-screen mode in web browsers.

### B. Alphanumeric Keys

These include:

- Letters A–Z.
- Numbers 0–9.
- Punctuation marks.
- Special characters.

They are mainly used for typing.

### C. Numeric Keypad

The numeric keypad is usually located on the right side of a full-size keyboard.
It contains:

- Numbers 0–9.
- Addition (+).
- Subtraction (-).
- Multiplication (*).
- Division (/).
- Decimal point.
- Enter.

The Num Lock key controls the numeric keypad's mode on many keyboards.

### D. Navigation Keys

These keys help users move around documents and screens.
Examples:

- Arrow keys.
- Home.
- End.
- Page Up.
- Page Down.

### E. Special Keys

These include:

**1. Enter:** Executes commands or moves the cursor to a new line.
**2. Spacebar:** Creates spaces between words.
**3. Backspace:** Deletes characters to the left of the cursor.
**4. Delete:** Deletes selected items or characters to the right of the cursor.
**5. Shift:** Allows users to type uppercase letters and access upper symbols on keys.
**6. Caps Lock:** Toggles uppercase letter typing.
**7. Tab:** Moves the cursor to the next tab stop or moves focus between controls.
**8. Esc:** Cancels or exits certain operations.
**9. Ctrl:** Used with other keys to perform shortcuts.
**10. Alt:** Used with other keys to access commands and system functions.
**11. Windows Key:** Opens the Start menu or works with other keys to perform Windows shortcuts.
**12. Fn:** Provides additional functions on many laptop keyboards.

## 15.4 Common Keyboard Shortcuts

Keyboard shortcuts allow users to perform tasks faster without navigating through menus.

| Shortcut | Function |
|---|---|
| Ctrl + A | Select all |
| Ctrl + B | Bold selected text |
| Ctrl + C | Copy |
| Ctrl + D | Function depends on the application |
| Ctrl + E | Centre alignment in many word processors |
| Ctrl + F | Find |
| Ctrl + H | Find and replace in many applications |
| Ctrl + I | Italic |
| Ctrl + J | Justify text in many word processors |
| Ctrl + K | Insert hyperlink in many applications |
| Ctrl + L | Left alignment in many word processors |
| Ctrl + M | Increase paragraph indentation in Word |
| Ctrl + N | Create a new document or window |
| Ctrl + O | Open a file |
| Ctrl + P | Print |
| Ctrl + R | Right alignment in many word processors |
| Ctrl + S | Save |
| Ctrl + T | Opens a new browser tab in most browsers |
| Ctrl + U | Underline |
| Ctrl + V | Paste |
| Ctrl + W | Close current document or tab |
| Ctrl + X | Cut |
| Ctrl + Y | Redo |
| Ctrl + Z | Undo |
| Alt + F4 | Close the active application |
| Alt + Tab | Switch between open applications |
| Windows + D | Show or hide the desktop |
| Windows + E | Open File Explorer |
| Windows + L | Lock the computer |

| | |
|---|---|
| Windows + I | Open Settings |
| Windows + R | Open the Run dialogue |
| Windows + Shift + S | Capture a selected screenshot area |
| Ctrl + Shift + Esc | Open Task Manager |
| Ctrl + Alt + Delete | Open Windows security options |

**Note:** Some shortcuts perform different functions depending on the application.

### 15.5 Cursor Movement and Text Selection

- Arrow keys: Move the cursor one character or line at a time.
- Home: Move to the beginning of the current line.
- End: Move to the end of the current line.
- Ctrl + Home: Move to the beginning of a document.
- Ctrl + End: Move to the end of a document.
- Page Up: Move up through a document.
- Page Down: Move down through a document.
- Shift + Arrow keys: Select characters or lines.
- Ctrl + Shift + Arrow keys: Select text word by word.
- Shift + Home: Select text from the cursor to the beginning of the line.
- Shift + End: Select text from the cursor to the end of the line.

### 15.6 Practical Keyboard Exercises

Students should practise:

1. Typing their full names.
2. Typing numbers from 1 to 100.
3. Typing a paragraph.
4. Using Caps Lock and Shift.
5. Copying and pasting text.
6. Selecting entire paragraphs.
7. Using Backspace and Delete.
8. Saving a document using Ctrl + S.
9. Switching between applications using Alt + Tab.

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## TOPIC 16: THE COMPUTER MOUSE AND POINTING DEVICES

### 16.1 Definition

A computer mouse is a pointing device used to control the movement of a pointer on a computer screen.

It enables users to select, open, move and interact with graphical elements.

### 16.2 Parts of a Computer Mouse

1. Left button.
2. Right button.
3. Scroll wheel.
4. Optical sensor.
5. USB cable or wireless receiver, depending on the model.

### 16.3 Types of Mouse

- Wired mouse.

- Wireless mouse.
- Optical mouse.
- Laser mouse.
- Gaming mouse.
- Ergonomic mouse.

## 16.4 Mouse Operations

### A. Pointing

Moving the mouse to position the pointer over an item on the screen.

### B. Single Clicking

Pressing and releasing the left mouse button once.
Uses:

- Selecting files.
- Selecting icons.
- Positioning the cursor.

### C. Double Clicking

Pressing the left mouse button twice quickly.
Uses:

- Opening folders.
- Launching desktop applications.
- Opening files.

### D. Right Clicking

Pressing the right mouse button once.
It usually displays a context menu containing additional commands.

### E. Dragging and Dropping

Dragging involves holding down the left mouse button while moving an item from one location to another.
Dropping involves releasing the mouse button at the desired destination.
Uses:

- Moving files.
- Repositioning windows.
- Moving objects in applications.

### F. Scrolling

Using the scroll wheel to move through documents and webpages.

## 16.5 Other Pointing Devices

1. Touchpad.
2. Trackball.
3. Joystick.
4. Stylus.
5. Touchscreen.
6. Graphics tablet.

## 16.6 Practical Mouse Exercises

Students should practise:

1. Moving the pointer around the screen.
2. Single-clicking desktop icons.
3. Double-clicking to open folders.

4. Right-clicking to display context menus.
5. Dragging and dropping icons.
6. Scrolling through a document.
7. Selecting multiple files.

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# TOPIC 17: INTRODUCTION TO THE WINDOWS OPERATING SYSTEM

## 17.1 What is Windows?

Windows is a graphical operating system developed by Microsoft.
It provides an environment for users to manage files, install and run applications, configure computer settings and interact with hardware.
Common versions include:

- Windows 10.
- Windows 11.

**Note:** Windows 10 reached the end of standard support on 14 October 2025. Windows 11 is the recommended current version for supported personal computers.

## 17.2 Main Components of the Windows Desktop

### A. Desktop

The desktop is the main workspace displayed after signing in.
It may contain:

- Icons.
- Shortcuts.
- Taskbar.
- Background image.

### B. Start Menu

The Start menu provides access to:

- Installed applications.
- Power options.
- User account options.
- Search.
- System settings.

**Procedure for opening the Start Menu:**

1. Click the Start button on the taskbar.
2. Alternatively, press the Windows key on the keyboard.

### C. Taskbar

The taskbar is normally located at the bottom of the Windows desktop.
It provides access to:

- Start menu.
- Search.
- Pinned applications.
- Running applications.
- System tray.
- Date and time.

### D. Search

Windows Search helps users locate applications, files, folders and system settings.

**Procedure:**

1. Click the Search icon or search box on the taskbar.
2. Type the name of the application or file.
3. Wait for the search results.
4. Select the required item.

### E. Notification Area/System Tray

This section displays system status icons such as:

- Network connection.
- Volume.
- Battery.
- Background applications.
- Security notifications.

### F. Recycle Bin

The Recycle Bin temporarily stores files deleted from local storage using normal deletion procedures.

It allows users to restore deleted files before they are permanently removed.

### G. File Explorer

File Explorer is the Windows application used to browse, organise and manage files, folders and storage devices.

Shortcut: `Windows + E`.

---

# TOPIC 18: WORKING WITH WINDOWS AND DESKTOP INTERFACE

## 18.1 Understanding a Window

A window is a rectangular area on the screen through which an application, folder or document is displayed.

Examples:

- Microsoft Word window.
- File Explorer window.
- Settings window.
- Google Chrome window.

## 18.2 Common Window Components

### A. Title Bar

Displays the name of the active application or document in applications that use a title bar.

### B. Minimise Button

Reduces a window to the taskbar without closing the application.

### C. Maximise Button

Expands a window to fill the available screen area.

### D. Restore Button

Returns a maximised window to its previous size.

### E. Close Button

Closes the active window or application.

### F. Scroll Bars
Allow users to move through content that cannot fit within the visible area.
### G. Menu Bar
Contains menus that provide access to commands in applications that use traditional menu bars.
### H. Toolbar
Contains buttons or tools used to perform common operations.
### I. Status Bar
Displays information about the current application or document.
**Note:** Not every modern Windows application contains all these components. Their arrangement depends on the application.

## 18.3 Window Management Procedures
**To minimise a window:**
1. Open an application.
2. Click the Minimise button.
3. The application remains open on the taskbar.

**To maximise a window:**
1. Open the application.
2. Click the Maximise button.
3. The window expands to occupy the screen.

**To restore a window:**
1. Open a maximised window.
2. Click the Restore Down button.

**To close a window:**
1. Click the Close button.
2. Alternatively, press Alt + F4.

## 18.4 Changing the Desktop Background
**Procedure in Windows 11:**
1. Right-click an empty area of the desktop.
2. Select Personalize.
3. Select Background.
4. Choose the desired background type.
5. Select Picture, Solid colour or Slideshow.
6. Browse and select a picture if necessary.
7. Adjust the picture display option.
8. Close Settings when finished.

## 18.5 Changing the Date and Time
**Procedure:**
1. Click Start.
2. Open Settings.
3. Select Time & language.
4. Click Date & time.
5. Configure the time zone.
6. Enable automatic time setting where appropriate.
7. If manual adjustment is required, turn off automatic time setting.
8. Click Change beside the manual date and time option.

9. Enter the correct date and time.
10. Save the changes.

For computers in Kenya, the appropriate time zone is East Africa Time (UTC+03:00).

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# TOPIC 19: WORKING WITH FILES AND FOLDERS

## 19.1 Understanding Files

A file is a named collection of digital information stored on a computer or another storage device.

Examples:

- Report.docx
- Students.xlsx
- Logo.png
- Notes.pdf
- Video.mp4

## 19.2 File Extensions

A file extension is the suffix at the end of a filename that indicates the file type or format.

Examples:

| Extension | File Type |
|---|---|
| .docx | Microsoft Word document |
| .xlsx | Microsoft Excel workbook |
| .pptx | PowerPoint presentation |
| .pdf | Portable Document Format |
| .txt | Plain text document |
| .jpg | Image |
| .png | Image |
| .mp3 | Audio |
| .mp4 | Video |
| .zip | Compressed archive |
| .html | HTML webpage |

## 19.3 Understanding Folders

A folder is a container used to organise files and other folders.

A folder created inside another folder is called a subfolder.

### Example of Folder Organisation

ICT ESSENTIALS

- WEEK ONE
  - Introduction to Computers.docx
  - Computer Hardware.pdf
- WEEK TWO
  - Windows Operations.docx
  - Practical Exercises.docx
- ASSIGNMENTS
  - Assignment One.docx

- Assignment Two.docx

Organising files properly makes it easier to retrieve information.

## 19.4 Creating a Folder

**Method One: Using the Mouse**

1. Open File Explorer.
2. Navigate to the location where you want to create the folder.
3. Right-click an empty area.
4. Select New.
5. Click Folder.
6. Type the folder name.
7. Press Enter.

**Method Two: Using the Keyboard**

1. Open File Explorer.
2. Navigate to the required location.
3. Press `Ctrl + Shift + N`.
4. Type the folder name.
5. Press Enter.

## 19.5 Creating a Subfolder

1. Open File Explorer.
2. Locate the main folder.
3. Double-click to open it.
4. Right-click an empty area.
5. Select New.
6. Click Folder.
7. Enter the subfolder name.
8. Press Enter.

## 19.6 Renaming a File or Folder

**Method One:**

1. Locate the file or folder.
2. Right-click it.
3. Select Rename.
4. Type the new name.
5. Press Enter.

**Method Two:**

1. Select the file or folder.
2. Press F2.
3. Enter the new name.
4. Press Enter.

## 19.7 Copying Files and Folders

Copying creates another instance of a file or folder while keeping the original.

**Procedure:**

1. Open File Explorer.
2. Locate the file or folder.
3. Right-click it.
4. Select Copy.
5. Navigate to the destination folder.

6. Right-click an empty area.
7. Select Paste.

**Keyboard shortcuts:**

- Ctrl + C: Copy.
- Ctrl + V: Paste.

## 19.8 Moving Files and Folders

Moving transfers a file or folder from one location to another.

**Procedure:**

1. Locate the required file.
2. Right-click it.
3. Select Cut.
4. Open the destination folder.
5. Right-click an empty area.
6. Select Paste.

**Keyboard shortcuts:**

- Ctrl + X: Cut.
- Ctrl + V: Paste.

## 19.9 Deleting Files and Folders

**Procedure:**

1. Open File Explorer.
2. Locate the file or folder.
3. Select it.
4. Press Delete.
5. Confirm if prompted.

Alternatively, right-click and select Delete.

The item will normally be moved to the Recycle Bin when deleted from a local drive.

## 19.10 Restoring Deleted Files

1. Open the Recycle Bin.
2. Locate the deleted file or folder.
3. Right-click the item.
4. Select Restore.
5. The item returns to its original location, if that location is available.

## 19.11 Permanently Deleting Files

**Method One: From the Recycle Bin**

1. Open Recycle Bin.
2. Locate the item.
3. Right-click it.
4. Select Delete.
5. Confirm the action.

**Method Two: Using the Keyboard**

1. Select the file or folder.
2. Press `Shift + Delete`.
3. Confirm the operation.

**Warning:** Permanently deleted files may be difficult or impossible to recover. Always confirm that the correct file is selected before performing this operation.

## 19.12 Searching for Files and Folders

1. Open File Explorer.
2. Navigate to the location where you expect the file to be.
3. Click the search box.
4. Enter the file or folder name.
5. Wait for the results.
6. Open the required item.

### 19.13 Creating a Text Document

A text document is a digital file used to store written information.

**Procedure for creating a text document in Windows:**

1. Open File Explorer.
2. Navigate to the desired location.
3. Right-click an empty area.
4. Select New.
5. Select Text Document.
6. Enter the filename.
7. Press Enter.
8. Double-click the file to open it in Notepad.
9. Type the required information.
10. Press Ctrl + S to save.

### 19.14 Creating a New Folder Structure for Students

Students should practise creating the following structure:

**ICT ESSENTIALS**

- WEEK 1
  - Theory Notes
  - Practical Work
- WEEK 2
  - Theory Notes
  - Practical Work
- ASSIGNMENTS
- PERSONAL DOCUMENTS

This exercise introduces proper digital file organisation.

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## TOPIC 20: BASIC COMPUTER OPERATIONS AND PROCEDURES

### 20.1 Opening an Application

Applications can be opened using several methods.

**Method One: Start Menu**

1. Click Start.
2. Search for the application name.
3. Click the application in the results.

**Method Two: Desktop Shortcut**

1. Locate the application's desktop icon.
2. Double-click the icon.

**Method Three: Taskbar**

1. Locate the application icon on the taskbar.
2. Click it.

**Method Four: Run Command**

1. Press Windows + R.
2. Type the appropriate application command.
3. Press Enter.

Examples:

- `notepad` opens Notepad.
- `calc` opens Calculator.
- `mspaint` opens Microsoft Paint, where available.

## 20.2 Closing an Application

1. Click the Close button.
2. Alternatively, press Alt + F4.
3. If prompted to save changes, choose Save, Don't Save or Cancel as appropriate.

## 20.3 Saving a Document

Saving stores the work created in an application.

**Procedure:**

1. Open the document.
2. Click File.
3. Select Save As if saving for the first time or choosing another location.
4. Browse to the desired folder.
5. Enter a suitable filename.
6. Select the required file format.
7. Click Save.

**Keyboard shortcuts:**

- Ctrl + S: Save.
- F12: Opens Save As in Microsoft Word and some other Office applications.

## 20.4 Opening an Existing Document

1. Open the appropriate application.
2. Click File.
3. Select Open.
4. Browse to the location of the document.
5. Select the file.
6. Click Open.

Alternatively, double-click the file in File Explorer if the correct application is associated with its file type.

## 20.5 Copying Text Between Documents

1. Open the source document.
2. Highlight the required text.
3. Press Ctrl + C.
4. Open the destination document.
5. Position the cursor where the text should appear.
6. Press Ctrl + V.
7. Save the destination document.

## 20.6 Connecting a USB Flash Drive

1. Identify an available USB port.

2. Insert the flash drive carefully.
3. Wait for Windows to recognise the device.
4. Open File Explorer.
5. Select This PC.
6. Locate the flash drive under Devices and drives.
7. Double-click to open it.
8. Copy or move files as required.

**Safely removing a flash drive:**

1. Close files and folders currently using the flash drive.
2. Locate the Safely Remove Hardware option in the system tray, where available.
3. Select the flash drive.
4. Wait for confirmation that it is safe to remove.
5. Disconnect the device.

Alternatively, use the Eject option in File Explorer where available.

## 20.7 Connecting to a Wi-Fi Network

1. Click the Network icon on the taskbar.
2. Ensure Wi-Fi is switched on.
3. Select the required wireless network.
4. Click Connect.
5. Enter the network password.
6. Click Next or Connect.
7. Wait for the connection to be established.

Only connect to networks you are authorised to use.

## 20.8 Taking a Screenshot

A screenshot is an image capture of the content displayed on a computer screen.

Common methods in Windows:

- `PrtScn`: Captures the screen or copies it to the clipboard, depending on the configuration.
- `Windows + PrtScn`: Saves a screenshot to the Screenshots folder on supported systems.
- `Windows + Shift + S`: Opens the screen snipping interface.
- `Alt + PrtScn`: Captures the active window on supported configurations.

## 20.9 Checking Computer Specifications

Students should know how to identify basic computer specifications.

**Method One: Settings**

1. Click Start.
2. Open Settings.
3. Select System.
4. Scroll down and select About.
5. Identify:
    - Processor.
    - Installed RAM.
    - Device name.
    - Windows edition.
    - System type.

**Method Two: Task Manager**

1. Press Ctrl + Shift + Esc.
2. Open the Performance tab.

3. Select CPU to view processor information.
4. Select Memory to view RAM information.
5. Select Disk to view storage activity and drive details.
6. Select Wi-Fi or Ethernet to view network activity, where available.

**Method Three: System Information**

1. Press Windows + R.
2. Type `msinfo32`.
3. Press Enter.
4. View detailed system information.

### 20.10 Basic Printer Setup

1. Connect the printer to the computer using USB or an available network connection.
2. Switch on the printer.
3. Open Settings.
4. Select Bluetooth & devices.
5. Select Printers & scanners.
6. Click Add device.
7. Wait for Windows to detect the printer.
8. Select the printer.
9. Follow the installation instructions.
10. Print a test page where the option is available.

Some printers require manufacturer-specific drivers or installation software.

# TOPIC 21: COMPUTER LABORATORY SAFETY AND MAINTENANCE

### 21.1 Definition of a Computer Laboratory

A computer laboratory is a room or designated facility equipped with computers and related resources for computer-based learning, research, training and practical activities.

### 21.2 Importance of Computer Laboratory Safety

Laboratory safety protects:

- Students.
- Trainers.
- Computer equipment.
- Electrical installations.
- Digital information.
- The learning environment.

### 21.3 Physical Safety Measures

#### 1. Proper Sitting Posture

Students should maintain proper posture when using computers.
Recommendations:

- Sit upright.
- Keep feet flat on the floor or a footrest.
- Position the monitor at a comfortable viewing height.
- Keep wrists in a relaxed position.
- Avoid bending excessively towards the screen.

### 2. Proper Ventilation
Computer laboratories should have adequate airflow to prevent overheating.

### 3. Electrical Safety
- Avoid touching electrical equipment with wet hands.
- Do not overload sockets.
- Use properly rated power extensions.
- Report damaged electrical cables.
- Avoid unauthorised electrical repairs.

### 4. Cable Management
Arrange cables properly to prevent accidents and accidental disconnections.

### 5. Cleanliness
- Keep workstations clean.
- Avoid eating and drinking near computers.
- Keep dust away from equipment.
- Clean keyboards and mice appropriately.

### 6. Fire Safety
- Install suitable fire extinguishers.
- Keep emergency exits accessible.
- Train staff and students on emergency procedures.
- Report electrical burning smells immediately.

### 7. Proper Handling of Equipment
- Do not drop computer equipment.
- Avoid pulling cables forcefully.
- Handle storage devices carefully.
- Do not disconnect equipment while it is in use.

### 8. Regular Breaks
Students should take regular breaks to reduce eye strain and physical discomfort.
A useful practice is the 20-20-20 rule: every 20 minutes, look at something approximately 20 feet away for at least 20 seconds.

## 21.4 Computer Maintenance
Computer maintenance involves activities performed to keep computer systems functional, secure and efficient.

### A. Preventive Maintenance
Preventive maintenance involves carrying out regular activities to reduce the likelihood of equipment failure.
Examples:
- Cleaning computer equipment.
- Updating software.
- Checking cables.
- Running security scans.
- Backing up important information.
- Monitoring storage capacity.

### B. Corrective Maintenance
Corrective maintenance involves repairing faults after they have occurred.
Examples:

- Replacing a damaged keyboard.
- Repairing a faulty power supply.
- Reinstalling corrupted software.
- Replacing a damaged storage drive.

### C. Software Maintenance

Includes:

- Updating applications.
- Updating operating systems.
- Removing unnecessary software.
- Scanning for malware.
- Managing storage space.

### D. Hardware Maintenance

Includes:

- Cleaning fans and vents.
- Inspecting cables.
- Checking power connections.
- Replacing faulty components.
- Ensuring adequate cooling.

## 21.5 General Laboratory Rules

1. Enter the computer laboratory only when authorised.
2. Do not eat or drink near computer equipment.
3. Do not install unauthorised software.
4. Do not change computer configurations without permission.
5. Do not share passwords.
6. Do not access another student's files without permission.
7. Report faulty equipment immediately.
8. Save work regularly.
9. Shut down computers properly.
10. Leave workstations clean and organised.

# TOPIC 22: DATA SECURITY AND COMPUTER PROTECTION

## 22.1 Introduction to Data Security

Data security refers to the practices, technologies and procedures used to protect digital information from unauthorised access, alteration, disclosure, destruction or loss.

Data security is important to individuals, businesses, educational institutions and government organisations.

Computers store sensitive information such as:

- Personal details.
- Student records.
- Examination results.
- Financial information.
- Passwords.
- Business records.
- Medical records.

Such information must be properly protected.

## 22.2 Objectives of Data Security

Data security is commonly based on three major principles known as the CIA Triad.

### A. Confidentiality

Confidentiality ensures that information is accessed only by authorised individuals.
Example: Only authorised college administrators should access confidential student records.

### B. Integrity

Integrity ensures that information remains accurate, complete and protected against unauthorised modification.
Example: Examination marks should not be changed by unauthorised users.

### C. Availability

Availability ensures that authorised users can access information and computer systems when required.
Example: A college should maintain reliable backups so that student records can be restored when a computer fails.

## 22.3 Common Computer Security Threats

### 1. Computer Viruses

A computer virus is malicious software that attaches itself to legitimate files or programs and can spread when infected files are executed.
It may:

- Corrupt files.
- Interfere with computer operations.
- Spread to other files.
- Cause data loss.

### 2. Worms

A worm is malicious software that can replicate and spread between computers, often through networks, without requiring a user to manually copy an infected file.

### 3. Trojan Horses

A Trojan horse is malicious software disguised as a legitimate or useful program.
It may allow attackers to access information or perform unauthorised activities.

### 4. Ransomware

Ransomware is malicious software that locks or encrypts files and demands payment for their restoration.

### 5. Spyware

Spyware secretly monitors user activities and may collect personal or confidential information.

### 6. Phishing

Phishing is a cyberattack in which criminals use fraudulent emails, messages, websites or phone communications to trick users into revealing confidential information.

### 7. Social Engineering

Social engineering involves manipulating people into revealing sensitive information or performing actions that compromise security.

### 8. Unauthorised Access

This occurs when someone accesses a computer, network, application or file without permission.

### 9. Data Theft

Data theft involves illegally obtaining or copying confidential information.

### 10. Physical Theft

Theft of computers, phones and storage devices can expose sensitive information stored on them.

## 22.4 Common Signs of a Compromised Computer

Possible warning signs include:

1. Unexpected pop-up messages.
2. Unusual computer behaviour.
3. Unknown applications appearing.
4. Unauthorised changes to files.
5. Unusual network activity.
6. Unexpected browser redirects.
7. Security software being disabled unexpectedly.
8. Unfamiliar login notifications.
9. Files becoming inaccessible or renamed.
10. Unusual messages being sent from the user's accounts.

**Important:** These symptoms do not always mean a computer has malware. They should be investigated rather than treated as automatic proof of infection.

## 22.5 Methods of Malware Transmission

Malware can spread through:

1. Infected USB flash drives.
2. Malicious email attachments.
3. Unsafe websites.
4. Untrusted software downloads.
5. Compromised applications.
6. Malicious links.
7. Infected network-connected devices.
8. Unauthorised software installations.

## 22.6 Methods of Protecting Computer Systems

### A. Use Antivirus and Endpoint Protection Software

Antivirus software helps detect, block and remove malicious software.

Examples:

- Microsoft Defender Antivirus.
- Bitdefender.
- ESET.
- Kaspersky.
- Avast.

**Procedure for scanning a computer using Microsoft Defender:**

1. Click Start.
2. Open Settings.
3. Select Privacy & security.
4. Click Windows Security.
5. Select Virus & threat protection.
6. Click Quick scan.
7. Allow the scan to complete.
8. Review the results.
9. Follow the recommended actions if threats are detected.

For a more comprehensive examination, select Scan options and choose Full scan.

### B. Use Strong Passwords

A password is a secret combination of characters used to authenticate a user.
A strong password should:

- Be sufficiently long.
- Be difficult to guess.
- Avoid personal information.
- Not be reused across important accounts.
- Be kept confidential.

A password manager can help users generate and securely store unique passwords.

## C. Enable Multi-Factor Authentication (MFA)

Multi-factor authentication provides additional protection by requiring more than one form of verification.
Examples:

- Password plus authentication application.
- Password plus security key.
- Password plus biometric verification.

## D. Keep Software Updated

Software updates help correct security weaknesses and improve system performance.
**Procedure for updating Windows 11:**

1. Click Start.
2. Open Settings.
3. Select Windows Update.
4. Click Check for updates.
5. Allow Windows to search for available updates.
6. Download and install the required updates.
7. Restart the computer if requested.

Applications should also be updated regularly.

## E. Avoid Suspicious Links and Attachments

Before opening links or attachments:

- Verify the sender.
- Check the website address.
- Avoid unexpected attachments.
- Do not enter passwords on suspicious websites.
- Do not download unknown programs.

## F. Use Secure Networks

- Avoid sharing sensitive information over untrusted public Wi-Fi.
- Use secure Wi-Fi passwords.
- Enable network security features.
- Keep routers and other network devices updated.

## G. Control Access to Computer Systems

- Use individual user accounts.
- Avoid sharing login credentials.
- Lock computers when leaving workstations.
- Restrict access to confidential information.
- Use appropriate user permissions.

## H. Protect Physical Devices

- Keep computers in secure locations.

- Use device locks where appropriate.
- Do not leave flash drives unattended.
- Avoid exposing devices to water, heat and physical damage.

## 22.7 Data Backup

Data backup is the process of creating additional copies of information so that it can be recovered in case the original data is lost, damaged or corrupted.

### Importance of Data Backup

1. Prevents permanent data loss.
2. Supports recovery after computer failure.
3. Protects against ransomware.
4. Supports business continuity.
5. Helps recover accidentally deleted information.

### Types of Data Backup

**1. Full Backup**

A full backup copies all selected data.

**2. Incremental Backup**

An incremental backup copies data that has changed since the most recent backup.

**3. Differential Backup**

A differential backup copies data that has changed since the last full backup.

### Backup Storage Locations

- External hard drives.
- USB storage devices.
- Network-attached storage.
- Cloud storage.
- Other secure storage systems.

### The 3-2-1 Backup Rule

A useful backup strategy is the 3-2-1 rule:

- Maintain at least three copies of important data.
- Store the copies on two different types of storage media.
- Keep at least one copy offsite or in a separate secure location.

### Procedure for Backing Up Important Files

1. Identify the files that require protection.
2. Organise the files into appropriate folders.
3. Connect an external storage device or access an approved cloud storage account.
4. Copy the files to the backup destination.
5. Verify that the files have been copied successfully.
6. Safely disconnect the external device where applicable.
7. Store the backup securely.
8. Repeat the process regularly.

**Important:** A backup should be tested periodically to confirm that files can actually be restored.

## 22.8 Safe Use of USB Flash Drives

USB flash drives can transfer malware from one computer to another.

Students should observe the following:

1. Avoid using unknown flash drives.
2. Scan external storage devices before opening their contents.
3. Avoid opening suspicious executable files.

4. Do not use flash drives found in public places.
5. Keep important information backed up.
6. Safely eject flash drives before disconnecting them.
7. Avoid storing confidential information on unprotected removable devices.

## 22.9 Safe Internet Practices

1. Visit trusted websites.
2. Use secure websites when submitting sensitive information.
3. Avoid downloading pirated software.
4. Do not disclose passwords through email or messages.
5. Avoid suspicious online advertisements.
6. Verify information before sharing it.
7. Log out of accounts when using shared computers.
8. Avoid saving passwords on public computers.
9. Be cautious when using public Wi-Fi.
10. Keep browsers and applications updated.

## 22.10 Protecting Personal and Institutional Information

Students and staff must observe responsible information-handling practices.
For an educational institution:

- Student records should only be accessed by authorised personnel.
- Examination records should be protected against unauthorised modification.
- Financial information should be kept confidential.
- Personal information should not be shared without proper authorisation.
- Important institutional records should be backed up.
- Computers containing confidential information should be secured.
- Unused user accounts should be disabled or removed according to institutional procedures.

## 22.11 What to Do When a Computer Is Suspected of Being Infected

1. Stop using the affected computer for sensitive activities.
2. Disconnect it from the network if there is a suspected active infection or unauthorised access.
3. Do not delete random system files.
4. Notify the computer laboratory instructor or IT administrator.
5. Run an appropriate security scan using trusted security software.
6. Follow the recommended malware removal procedures.
7. Change potentially compromised passwords using a clean, trusted device.
8. Restore affected files from verified backups if necessary.
9. Update the operating system and applications.
10. Investigate how the infection occurred to prevent recurrence.

**Important:** In an institutional environment, students should not attempt major system repairs or reinstallations without authorisation.

# WEEK ONE AND TWO: STUDENT PRACTICAL ASSESSMENT

Students should be able to demonstrate the following skills before proceeding to the next section of ICT Essentials.

## Practical Exercise 1: Computer Identification

1. Identify the main external components of a desktop computer.
2. Identify at least five computer ports.
3. Explain the functions of the identified ports.
4. Identify the main components found inside a system unit using available demonstration equipment.
5. Differentiate between hardware and software.

## Practical Exercise 2: Computer Startup and Shutdown

1. Start a computer from a powered-off state.
2. Identify the Windows login screen.
3. Sign in using an authorised account.
4. Restart the computer correctly.
5. Put the computer into Sleep mode.
6. Shut down the computer properly.

## Practical Exercise 3: Keyboard and Mouse Operations

1. Type a short paragraph.
2. Use Caps Lock and Shift.
3. Demonstrate copying and pasting.
4. Use keyboard shortcuts to save a document.
5. Demonstrate single-clicking, double-clicking and right-clicking.
6. Demonstrate dragging and dropping.
7. Open two applications and switch between them.

## Practical Exercise 4: File and Folder Management

1. Open File Explorer.
2. Create a folder named ICT ESSENTIALS.
3. Create two subfolders named WEEK ONE and WEEK TWO.
4. Create a text document.
5. Rename the document.
6. Copy the document into another folder.
7. Move the document to a different location.
8. Delete the document.
9. Restore the deleted document from the Recycle Bin.

## Practical Exercise 5: Basic Windows Operations

1. Open Notepad.
2. Type a paragraph about the importance of computers.
3. Save the document as Introduction to Computers.txt.
4. Change the desktop background.
5. Open Windows Settings.
6. Identify the computer's RAM and processor.
7. Take a screenshot using Windows + Shift + S.
8. Connect to an authorised Wi-Fi network.

## Practical Exercise 6: Data Security

1. Open Windows Security.
2. Identify the Virus & threat protection section.
3. Perform a Quick scan.
4. Explain the importance of strong passwords.
5. Demonstrate how to safely remove a USB flash drive.
6. Create a backup folder on an external storage device.
7. Explain the 3-2-1 backup rule.
8. Identify at least five examples of cyber threats.

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# REVISION QUESTIONS

1. Define a computer.
2. Explain the four stages of the information processing cycle.
3. Differentiate between data and information.
4. Explain five characteristics of computers.
5. Classify computers according to size and functionality.
6. Explain five applications of computers in modern society.
7. Differentiate between hardware and software.
8. Explain the functions of the motherboard.
9. Identify and explain five input devices.
10. Identify and explain five output devices.
11. Differentiate between primary memory and secondary storage.
12. Explain the differences between RAM and ROM.
13. Differentiate between HDD and SSD.
14. Explain the functions of the CPU.
15. Describe the Fetch–Decode–Execute cycle.
16. Differentiate between system software and application software.
17. Explain five functions of an operating system.
18. Differentiate between cold booting and warm booting.
19. Identify the main sections of a computer keyboard.
20. Explain ten commonly used keyboard shortcuts.
21. Describe five mouse operations.
22. Identify the major components of the Windows desktop.
23. Explain the procedure for creating, renaming, copying and deleting folders.
24. Explain the difference between Sleep, Hibernate and Shut down.
25. Describe five computer laboratory safety measures.
26. Differentiate between preventive and corrective maintenance.
27. Explain the CIA Triad in data security.
28. Identify five common computer security threats.
29. Explain five methods of protecting computer systems.
30. Describe the 3-2-1 backup rule.

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**END OF SECTION**
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